

CRÎȘMARIU CODRIN

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EDUCATION

2024 - 2028 Tufts University **GPA:** 3.96

Pursing a double degree in Mechanical Engineering and Computer Science

EXPERIENCE

2026 - Present SPARC Lab Student Researcher

- Developing a hybrid control framework for the Unitree G1 humanoid robot to navigate sparse footholds by integrating Reinforcement Learning (PPO) with physics-based guides.
- Implementing and comparing three training architectures: a heuristic baseline, a Linear Inverted Pendulum (LIP) guided model, and a full-order Model Predictive Control (MPC) guided system.
- Utilizing Control Lyapunov Functions (CLF) within the RL reward structure to embed stability guarantees and physical consistency into the learning process.
- Simulating complex dynamics in MuJoCo to enable precise foot placement across both sagittal and frontal axes

2024 - 2025 **Tufts Center for Engineering Outreach** Student Researcher

- Developed a new embedded firmware for the OpenMV camera powered by the ESP32, capable of running lightweight face landmark detection on resource-constrained devices.

2024 - 2025 **Tufts Solar Vehicle Project** Mechanical Lead

- Designed the steering system of our competition car, working with custom carbon fiber composites and aluminum CNC machining.
- Lead a team of 10 students in designing a steel-welded rollcage and the roof angle-ing mechanism to maximize the amount of solar power the car gets while idle.

2020 - 2024 **FIRST Tech Challenge Robotics Team Peppers** Team Leader

- Managed a team of 20 members
- Managed and volunteered in over 10 team-organized events and helped in raising over \$114,000 through partnerships with companies such as Amazon, KPMG, Mercedes, and Pentalog
- Coordinated the 3D design, mechanical, and software team in creating the robot voted 2nd in the world
- Productionized over 6000 lines of code in Java. Utilized industry-standard control techniques and algorithms including inverse kinematics, feedforward/PIDF controllers, motion profiling, and computer vision

PERSONAL PROJECTS

HUMANOID FALL RECOVERY

Developed a robust control policy for a humanoid robot capable of recovering from falls and transitioning into a stable walking gait using curriculum learning. This project was built for the Reinforcement Learning class at Tufts University using Unity ML-Agents and Proximal Policy Optimization.

ZORO ROBOT DOG

Fully autonomous 3D-printed robotic dog for ISEF, designed to map and navigate its surroundings. Equipped with a modified Xbox Kinect for RGBD mapping, encoders for proprioception, high-torque servos, and an Intel NUC computing unit for mapping and localisation. I researched robot control theory, gait dynamics, and inverse kinematics, and designed all mechanical components in Fusion 360.

OPEN-DRONE

Open-source autonomous quadcopter with a custom cost-efficient stereo camera for Visual Stereo SLAM for the Romania National Olympiad of Innovation
Researched quadcopter kinematics, visual stereo simultaneous localization, and mapping using algorithms such as sparse optical flow and orb feature detection and matching

HONORS

ISEF GRAND AWARD WINNER • Won the grand award in the robotics section with a completely autonomous robot dog that could map and navigate its surroundings

FIRST TECH CHALLENGE • Second Place Motivate Award at the FIRST World Championship 2023
• Robot voted 2nd in the World in 2023
• Second Place Inspire Award at the FIRST Romanian National Championship 2023
• Third Place Think Award at the FIRST Romanian National Championship 2022

RISE FOR THE WORLD • Rise for the World finalist award, ranking in the top 500 of the 90,000 participants